1st Lab Assignment

Tools: Students' choice

Required Material:

1. Textbook.
2. Lecture Notes.
3. Lab worksheet.
4. Example video
   https://www.youtube.com/live/QM14OWuTYvk?si=MbVojBl9yI2NvT6w

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Tasks:

- Create a simple application using any programming language you prefer (HTML and Php if you want to follow the provided video)
- Perform at least two different SQL injections.
- Sanitize your code to prevent these SQL injections.

Deliverables:

- A report with all steps performed (Should be uploaded to the blackboard)
- Quick Presentation

Evaluation Rubric:

Criteria	Marks	Description
Application Setup	0.75	- Properly created a simple application (Php /Java/Python or any other language you choose) with a vulnerable login form. The application should: - Ask for some user input (ex: username + password) - lookup something in the database given the input. - Return some value (Ex: Username)
	0.25	- Correct database connection and table structure.
SQL Injection Attack #1	1	- Successfully performed SQL injection to bypass authentication. - Extract user data through experimenting with strings. - Clearly documented steps and SQL query generated.
SQL Injection Attack #2	1	- Successfully performed SQL injection to retrieve DB information or users data. - Clearly documented steps and SQL query generated.
Code Sanitization	1.5	- Implemented SQL injection prevention (At least 2 methods). - Explanation of how your solution is effective for securing input. - Demonstrated application security post-sanitization (screenshots).
Report Quality	0.5	- Clear and organized report. - Screenshots, explanations presented of each step.
Presentation	2.5	During the 6th lab